

Structural & Mechanism Design Document

ASEN 5148-001: Spacecraft Design

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1 Executive Summary

This report presents the design and structural analysis of a spacecraft bus developed to withstand launch loads while maintaining structural efficiency and supporting all onboard subsystems. The spacecraft is configured as a $3.3\text{ m} \times 1.8\text{ m} \times 1.8\text{ m}$ monolithic structure, interfacing with a 1575 mm Payload Adapter Fitting (PAF) and a 1194 mm clampband, which together form the primary load transfer path to the launch vehicle.

The design utilizes a hybrid material approach, combining Al 7075-T6 primary structural members with Duxford HRH honeycomb sandwich panels for external surfaces. This configuration provides an effective balance between strength, stiffness, and mass efficiency while remaining compatible with aerospace structural standards. The internal layout includes a propulsion deck supporting a high-mass propellant system, along with structural provisions for avionics, payload, and deployable solar arrays.

The design process involved defining system-level geometry, selecting materials based on performance requirements, and performing analytical calculations to evaluate structural behaviour under launch conditions. Based on the spacecraft wet mass of 1760.4 kg, quasi-static load cases derived from the Falcon 9 User's Guide for a medium-class payload were applied, including +8.5 g axial compression, -4.0 g axial tension, and $\pm 3.0\text{ g}$ lateral acceleration. These conditions resulted in a maximum axial compression load of approximately 146.79 kN, which was used to size the primary structural members and evaluate stress and stability.

Results from the analysis show that the primary structure operates well below material yield limits, with calculated compressive stresses of approximately 24.93 MPa in the main longerons compared to a yield strength of approximately 503 MPa for Al 7075-T6, resulting in a factor of safety greater than 20 under limit loading. Buckling analysis indicates that structural stability governs the design, with a critical buckling load of approximately 44 kN per longeron, providing a margin of safety greater than 1.7 under ultimate loading conditions.

The propulsion deck, supporting a propellant mass exceeding 1000 kg, experiences axial loads of approximately 87.9 kN under launch conditions and is reinforced using underlying structural members and stiffeners to ensure effective load transfer. The tank mounting system, secured using 12 M10 bolts, is designed to distribute loads with approximately 7.3 kN axial load per bolt, within acceptable preliminary limits. Additionally, the aluminium honeycomb panel system contributes to overall structural stiffness and provides efficient load distribution while maintaining low mass.

Overall, the design meets the required structural performance criteria with significant safety margins while maintaining a lightweight and efficient configuration. The results demonstrate that the proposed structure is capable of withstanding expected launch conditions and supporting mission operations. Further refinement through detailed finite element analysis, modal characterization, and structural testing is recommended to validate and optimize the design.

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3 Requirements

The Structures and Mechanisms (SAM) subsystem requirements define the structural, mechanical, and operational criteria that the spacecraft must satisfy to ensure mission success. These requirements are derived from launch vehicle constraints, subsystem integration needs, and mission performance objectives.

The structure must be capable of withstanding launch loads, supporting all onboard subsystems, maintaining geometric and thermal constraints, and enabling proper functionality of mechanisms such as the solar array drive assembly and antenna gimbals. All requirements are defined with traceability and are verified through inspection, analysis, or testing.

3.1 System Requirements (Level IV)

SAM	STRUCTURE AND MECHANISMS
SAM_01	The spacecraft and all subsystem interfaces (decks, compartments, etc.) shall meet vibration, shock, and load limits and factors as specified in the Falcon 9 User's Guide.
SAM_02	The spacecraft structure will interface with the Falcon 9's 1575 mm PAF (Payload Attachment Fitting) through the -Z-Panel.
SAM_03	The structure subsystem shall provide secure mounting interfaces for other subsystems, including, but not limited to, an instrument suite deck, a propulsion deck, and a CDH compartment attachment.
SAM_04	The spacecraft structure shall survive nominal launch/flight environments (as specified in the Falcon 9 User's Guide) and nominal mission conditions.
SAM_05	The solar array drive assembly shall be able to maneuver to an accuracy of 0.1 deg or better to optimize power generation.
SAM_06	The spacecraft's dimensions (with stowed solar arrays) shall fit within the Falcon 9 fairing envelope (as specified in the Falcon 9 User's Guide).
SAM_07	All SAM gimbaling systems (solar array drive assembly and HGA gimbal) shall operate within the operational temperature range of -30 to 65 deg C.
SAM_08	The HGA gimbal shall be able to maneuver to an accuracy of 0.1 deg or better to maintain communications with the NSN.
SAM_09	All SAM gimbaling systems (solar array drive assembly and HGA gimbal) shall configure themselves according to different spacecraft operational modes, such as, but not limited to, "Safe Mode" and "Terrain Avoidance Mode" shown in the FPDD.
SAM_10	The spacecraft structure shall provide clear FOVs for all components and instruments required by their specifications.

SAM_11	The spacecraft structure shall be designed in a way such that the CG migration over the course of the mission does not negate thruster control authority, as shown in the PDR Design Document.
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Table 1: Structure and mechanisms system requirements

3.2 SAM_01 – Structural Load and Environment Compliance

The spacecraft structure is required to withstand all launch-induced loads, including vibration, shock, and quasi-static accelerations, as defined in the Falcon 9 User’s Guide.

This requirement ensures that both the primary structure and all subsystem interfaces remain intact and functional during launch. The current design addresses this through the use of a primary load-bearing frame, which provides high strength and stiffness, combined with aluminium honeycomb panels that enhance overall rigidity and load distribution. Preliminary calculations under 8.5 g axial and 3.0 g lateral loads show that stresses remain well below material yield limits, with maximum compressive stress in the longerons of approximately 24.93 MPa compared to a yield strength of ~503 MPa, indicating compliance at the sizing level.

- Status: Satisfied (Preliminary)
- Verification: Testing (planned), Analysis (Future work)

3.3 SAM_02 – Launch Vehicle Interface Compatibility

The spacecraft must interface with the Falcon 9 1575 mm Payload Adapter Fitting (PAF) through a structural interface.

This requirement is satisfied by designing a reinforced base interface (Z-panel region) that connects the primary frame directly to the clamp band interface. The design ensures a continuous load path from the spacecraft structure into the launch vehicle while maintaining alignment and minimizing stress concentrations.

- Status: Satisfied
- Verification: Inspection

3.4 SAM_03 – Subsystem Mounting Interfaces

The structure must provide secure mounting interfaces for all subsystems, including propulsion, instrumentation, and avionics.

This is achieved by incorporating dedicated structural decks and mounting points within the primary frame. The propulsion deck is specifically reinforced to support high loads, while the instrument deck provides stable mounting for payloads. The structural layout ensures proper load transfer and avoids localized overstressing.

- Status: Satisfied (Preliminary)
- Verification: Inspection

3.5 SAM_04 – Launch and Mission Environment Survivability

The spacecraft structure must survive both launch conditions and nominal mission environments.

This includes maintaining structural integrity under dynamic loading and environmental conditions. The use of aerospace-grade materials such as Ti-6Al-4V and aluminium alloys, combined with conservative design margins, ensures durability and reliability throughout the mission. The structural configuration is designed to handle both transient launch loads and steady-state in-orbit conditions.

- Status: Satisfied (Preliminary)
- Verification: Testing (planned), Analysis

3.6 SAM_05 – Solar Array Drive Assembly Accuracy

The Solar Array Drive Assembly (SADA) must achieve a pointing accuracy of 0.1° or better to optimize power generation.

This requirement is addressed structurally by providing high-stiffness mounting interfaces and minimizing deflections in the supporting structure. The primary frame design reduces vibration and deformation that could otherwise impact pointing accuracy. However, detailed structural and control coupling analysis is required to fully validate this performance.

- Status: Partially Satisfied (Requires detailed analysis)
- Verification: Analysis

3.7 SAM_06 – Fairing Envelope Constraint

The spacecraft must fit within the Falcon 9 fairing envelope in its stowed configuration.

The current design adheres to this requirement by maintaining overall dimensions of $3.3\text{ m} \times 1.8\text{ m} \times 1.8\text{ m}$, including stowed solar arrays. The layout has been optimized to ensure all components remain within allowable volume limits without interference.

- Status: Satisfied
- Verification: Inspection

3.8 SAM_07 – Thermal Operating Range for Gimbals

All gimbaling systems must operate within a temperature range of -30°C to $+65^\circ\text{C}$.

This requirement is addressed through the selection of materials with stable thermal properties and by designing interfaces that minimize thermal-induced deformation. While structural provisions support this requirement, detailed thermal analysis is required to fully validate performance.

- Status: Partially Satisfied (Preliminary)
- Verification: Testing (planned), Analysis

3.9 SAM_08 – High Gain Antenna (HGA) Pointing Accuracy

The HGA gimbal must achieve a pointing accuracy of 0.1° or better to maintain communication.

The structure supports this requirement by ensuring rigid mounting and minimal deflection at the antenna interface. Structural stiffness reduces misalignment under load; however, final validation will depend on integrated structural and control system analysis.

- Status: Partially Satisfied (Requires detailed validation)
- Verification: Testing, Analysis

3.10 SAM_09 – Multi-Mode Operation Capability

The spacecraft must support multiple operational modes, including Safe Mode and Terrain Avoidance Mode, without compromising mechanical functionality.

The structural design ensures that gimbaling systems are mounted on stable platforms that allow full range of motion. The structure does not impose constraints that would limit operational flexibility.

- Status: Satisfied (Preliminary)
- Verification: Testing/Analysis

3.11 SAM_10 – Field of View (FOV) Clearance

The structure must provide clear fields of view for all instruments and components.

This requirement has been addressed through careful placement of structural members to avoid obstruction of sensors, antennas, and payload systems. The layout ensures that all critical components have unobstructed operational paths.

- Status: Satisfied (Preliminary)
- Verification: Inspection

3.12 SAM_11 – Center of Gravity (CG) Stability

The structure must ensure that centre-of-gravity shifts during the mission do not negatively affect thruster control authority.

This is addressed by strategic placement of heavy components, such as the propulsion system, near the central axis and by maintaining a balanced mass distribution. Further analysis is required to evaluate CG migration over the mission lifecycle.

- Status: Partially Satisfied (Requires detailed CG analysis)
- Verification: Analysis

4 Subsystem Design Overview

The Structures and Mechanisms (SAM) subsystem defines the mechanical architecture of the spacecraft and ensures structural integrity throughout launch and mission operations. The design is organized into key functional elements that collectively support load transfer, subsystem integration, and deployable mechanisms.

- **Primary Structure (Bus Frame):**
The main load-bearing framework composed of longerons and structural members, designed to withstand launch loads and provide the primary load path from the spacecraft to the launch vehicle interface.
- **Decks & Panels:**
Includes propulsion and instrument decks along with honeycomb sandwich panels, which provide mounting surfaces, structural stiffness, and enclosure while distributing loads across the structure.
- **Propulsion Integration:**
Structural integration of the propellant tank and associated hardware, ensuring efficient load transfer through reinforced deck structures and support members.
- **Solar Array System:**
Incorporates the Solar Array Drive Assembly (SADA), hinges, and deployment mechanisms, enabling controlled deployment and orientation for power generation.
- **Payload & Avionics Mounting:**
Provides structurally stiff and stable mounting interfaces for avionics and payload components, ensuring minimal deflection and maintaining alignment during operation.
- **Launch Vehicle Interface:**
Defines the connection to the Falcon 9 Payload Adapter Fitting (PAF) and clampband system, forming the primary load transfer interface during launch.

5 Primary Structure (Frame System)

The structural concept assumed is a box-type bus with a lower interface frame and PAF attachment at the base, a set of longitudinal load-carrying members running vertically, internal decking for propulsion and avionics, and external HRH honeycomb panels used primarily for enclosure, local stiffness, and distributed equipment mounting.

The structure consists of a tubular frame arranged in a rectangular configuration. Vertical and horizontal members form a rigid load path connecting the propulsion deck, instrument deck, and interface ring.

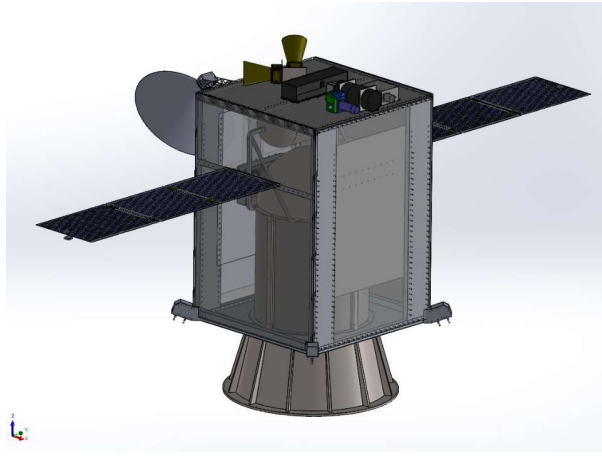


Figure 5.1: Primary Frame CAD Model

The primary load path is:

mounted equipment → local brackets/decks → longitudinal members and lower frame → PAF attachment → launch vehicle.

5.1 Honeycomb Panel System

The honeycomb panels serve both structural and protective functions, acting as shear walls while enclosing the spacecraft.

The panels are constructed as aluminium sandwich structures, consisting of:

- Outer face sheets: 0.8 mm CFRP
- Core: 19.8 mm HRH-10-3/16-5.4 honeycomb

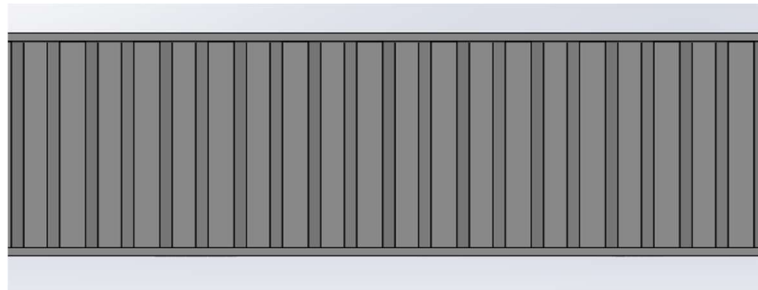


Figure 5.1: Panel Cross-Section

The use of sandwich panels provides:

- a) High bending stiffness with low weight
- b) Efficient shear load distribution
- c) Improved vibration resistance

Each panel contributes to overall structural stiffness and helps stabilize the frame under load conditions.

In this concept, external wall panels should not be treated as the only primary compression members. Their role should remain enclosure, shear stabilization, and localized support unless the panel inserts, close-outs, and corner joints are explicitly designed as a semi-monocoque load path and validated by analysis.

For the current structural configuration, the spacecraft utilizes aluminium honeycomb sandwich panels as the primary external enclosure. The panel geometry is defined based on the updated bus layout.

The top panel has dimensions of 1500 mm × 1500 mm, resulting in an area of:

$$A_{top} = 1.5 \times 1.5 = 2.25 \text{ m}^2$$

Each of the four side panels has dimensions of 2110 mm × 1492 mm, giving an area of:

$$A_{side} = 2.110 \times 1.492 = 3.15 \text{ m}^2 \text{ (per panel)}$$

For four side panels:

$$A_{sides,total} = 4 \times 3.15 = 12.60 \text{ m}^2$$

The total honeycomb panel surface area is therefore:

$$A_{total} = 2.25 + 12.60 = 14.85 \text{ m}^2$$

These panels act as both structural shear elements and protective enclosures, contributing to global stiffness while maintaining a low mass due to the honeycomb core construction.

5.2 Propulsion Deck Structure

The deck is constructed from Al 7075-T6 with a plate thickness of 2.5 mm, reinforced by 30 mm × 30 mm × 3 mm square tube stiffeners, and is supported by underlying structural beams. It supports a propulsion system mass of approximately 1054.8 kg (fuel/Pressurant), which governs the structural design of this subsystem. During launch, this results in axial loads of approximately 87.9 kN under 8.5 g loading, which are distributed into the vertical longerons and surrounding frame members.



(Figure 5.2: Propulsion Deck Layout)

Key design considerations:

- Reinforced mounting interface for tank support using a Ti-6Al-4V tank ring with 12 M10 bolts

- b) Central placement of the propulsion system to minimize centre of gravity (CG) offset
- c) Direct load transfer from the tank ring through the deck and into the primary structure

This design ensures that the propulsion system remains securely mounted and that loads are efficiently distributed into the primary structure without introducing excessive localized stresses.

5.3 Instrument Deck

The instrument deck provides mounting surfaces for payload and optics. It is designed to ensure stability and minimize deflection, which is critical for sensitive instruments.

The deck is structurally integrated into the upper portion of the frame and provides modular mounting interfaces for components.



(Insert Figure: Instrument Deck Layout)

Design features include:

- a) Rigid mounting surfaces for sensitive equipment
- b) Minimal structural deflection for accuracy
- c) Clear of obstruction for FOV
- d) Accessibility for integration and maintenance

This ensures proper functionality of instruments and maintains alignment requirements.

5.4 Avionics Integration

The avionics panel provides mounting surfaces for critical avionics components, including the Command and Data Handling (CDH), Guidance, Navigation and Control (GNC), and Telemetry (TELE) subsystems.

The panel is implemented as a hybrid structural assembly, consisting of the primary spacecraft structure integrated with a CFRP wall panel onto which the avionics boxes and associated components are mounted. The CFRP panel provides a lightweight and stiff mounting surface, while the underlying primary structure ensures effective load transfer into the spacecraft frame.

The assembly supports an avionics mass of approximately 54.4 kg, which results in axial loads of approximately 4.54 kN under 8.5 g launch conditions. These loads are distributed through the CFRP panel and transferred into the main structural members via bolted interfaces, ensuring structural integrity and minimizing local deformation.

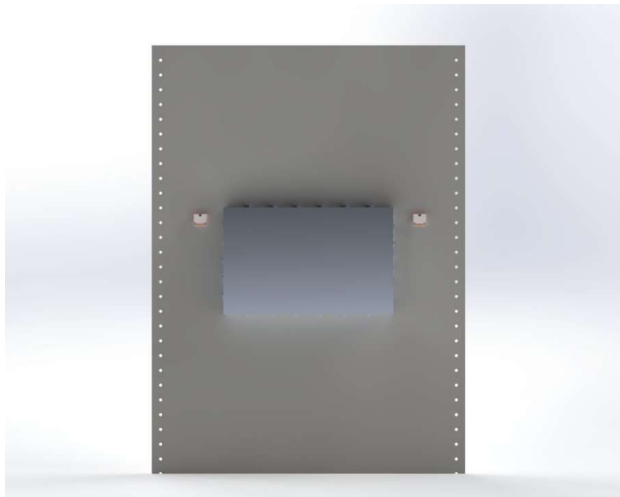


Figure 5.3: Avionics Panel / CFRP Wall Layout

The use of a CFRP panel allows for high stiffness-to-weight performance, reducing deflection and maintaining alignment of sensitive avionics components. The combined structure ensures secure mounting, efficient load transfer, and compatibility with the overall spacecraft structural system.

5.5 Launch Vehicle Interface

The spacecraft interfaces with the launch vehicle through a 1575 mm PAF and 1194 mm clampband system, which serves as the primary load transfer interface.

The interface is implemented through a reinforced base ring connected to the primary structure, ensuring a continuous load path.

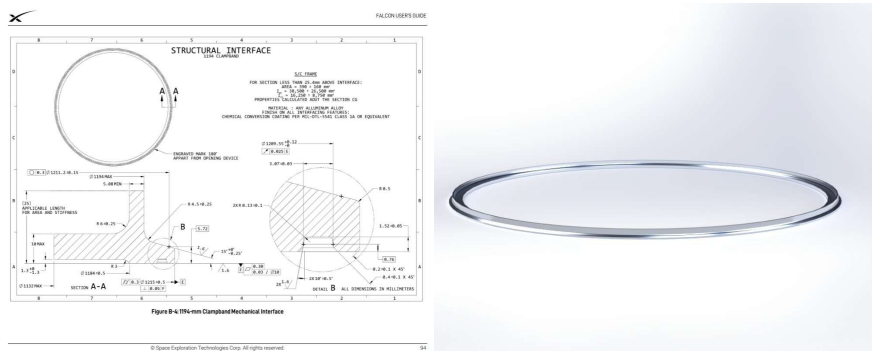


Figure 5.5: Clampband Interface

Important aspects of the design:

- a) Uniform load distribution around circumference

- b) Avoidance of stress concentrations
- c) Structural continuity between frame and interface

This configuration ensures that all launch loads are efficiently transferred to the launch vehicle during launch.

5.6 Solar Array Support and SADA Interface

The solar array system consists of two deployable arrays, each with three panels, mounted via a Solar Array Drive Assembly (SADA) and HDRMs.

The SADA is attached to a structurally reinforced interface to ensure minimal deflection and maintain pointing accuracy.

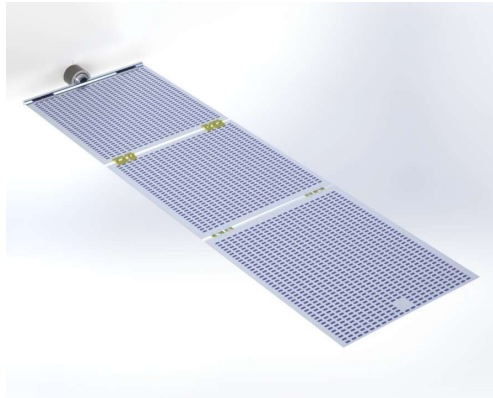


Figure 5.6: Solar Array and SADA

Design considerations:

- a) High stiffness to maintain 0.1° pointing accuracy
- b) Structural reinforcement at mounting interface
- c) Capability to withstand launch loads in stowed configuration

The structure supports both the mechanical loads and operational accuracy requirements of the solar array system.

5.7 HGA Mounting and Gimbal Support

The High Gain Antenna (HGA) requires precise positioning for communication. The structure provides a stable mounting interface to support the gimbal mechanism.

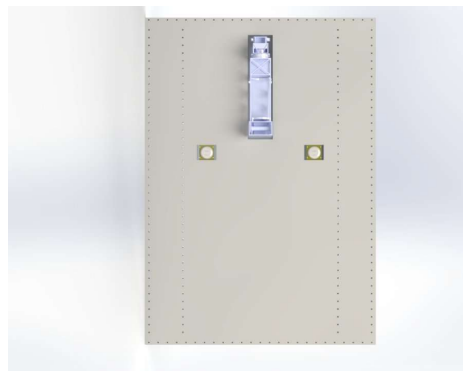


Figure 5.7: HGA Mount

Key features:

- a) Rigid support to minimize deflection
- b) Placement to avoid interference with other components
- c) Structural stability under load

This ensures that communication accuracy is maintained throughout operation.

5.8 Manufacturing and Assembly Considerations

The design has been developed with manufacturability and assembly in mind. Standard aerospace processes and materials are used to simplify fabrication.

- a) Frame: CNC machining and tube fabrication
- b) Panels: Bonded honeycomb construction
- c) Fasteners: Standard A286 bolts
- d) Assembly: Modular integration approach

This reduces complexity and improves build efficiency.

5.9 DFX Considerations

The design incorporates multiple Design-for-X (DFx) principles:

- a) DFM: Use of standard materials and simple geometries
- b) DFA: Modular design and accessible interfaces
- c) Reliability: Conservative safety margins and robust materials

These considerations improve manufacturability, assembly, and overall system reliability.

6 BOM (Bill of Materials)

Sub-Assembly No	Part No	Name	Description	Quantity
PROPULSION				
Prop 1		TCM system		
	P001	TCM base	Mounting for tcm	1
	P002	Lower support big	Interface between tcm and base	4
	P003	Lower support small	Interface between tcm and base	4
	P004	Upper lock big	Locks TCM in place	4
	P005	Upper lock small	Locks TCM in place	4
Prop 2	Prop Deck (RCS system)			
	PD001	Upper surface	Shell for the components	1
	PD002	Lower surface	Cover for the shell	1
	PD003	RCS Beam	Mounting surface and structure for the RCS	4
	PD004	RCS cover	Cover for the RCS system	4
	PD005	Base support beams	main structural members	4
Prop 3	Pressurant tank			
	PRT001	Side structs	Structure for the tank	2
	PRT002	Tank connector	To fix tanks position	2
	PRT003	Carbon tube	Structure support for upper frame	2
Prop 4	Prop tank			
	PT001	Prop Struct	structure for the prop tanks	4
	PT002	Tank support	Tank support	2
	PT003	Prop tank cover	To cover the tank	4

Sub-Assembly No	Part No	Name	Description	Quantity
STRUCTURE				
Struct1	Upper structure assembly			
	S004	Main beam	Load bearing beams	8
	S005	Base plate	Base to upper structure	4
	S006	Connecting struct	To assemble two sides of upper structure	4
	S007	Beam Cover	Wall covering and supporting main beams	4
Struct2	Top Enclosure assembly			
	S008	Main cross beam	Cross beam for structure	1
	S009	Sub beams	"+" beams for support	2
	S010	plate	to connect the smaller beams	1
	S011	Connectors L	L shaped connectors to connect the "+"	4
	S012	Bigger L connectors	To secure to the main structure	8
HONEYCOMB				
Top	HB01	face sheets		2
	HB05	HRH-honeycomb		1
Face1	HB02	face sheets		2
	HB06	HRH-honeycomb		1
Face2	HB03	face sheets		2
	HB07	HRH-honeycomb		1
Face3&4	HB04	face sheets		4
	HB08	HRH-honeycomb		2
FASTNERS				
	M4	To secure surfaces (With washers and nuts)		580
	M5			
	M6			
	M8			
	M10			

Sub-Assembly No	Part No	Name	Description	Quantity
AVIONICS & NAVIGATION				
CDH1	C001	Box base	Housing for CDH	1
	C002	Box top	Cover for CDH housing	1
	C003	Components Wall	Base wall on which components mount	1
TELE1	T001	HGA Base	Rigid structure for HGA+Gimbal	1
GNC1	G001	Wall	Base wall to mount RW	1
	G002	RW base	Structure to support RWs	2
	G003	STR base	Mounts for start trackers	1
INSTRUMENT				
INST1	I001	NFOV base	Payload dampeners and support	1
	I002	NFOV connector top	housing to secure payload	2
	I003	NFOV connector bottom	housing to secure payload	2
ELECTRICAL POWER SYSTEM				
EPS1	E001	Hinge	Hinge to deploy Solar arrays	4
	E002	Stopper	Hard stop to fic the max deployment	4
	E003	SADA	Solar array drive assembly	1
	E004	First hinge	Initial hinge between SADA and panels	1
	E005	SADA yoke	structure to mount panels	2
	E006	HDRM	To release the panels to be deployed	2
	E007	SADA struct	Structure to accommodate SADA into the BUS	2
	E008	SADA base mount	Structure to secure SADA into the BUS	2

7 Mass Breakdown and Structural Implications

The spreadsheet gives the subsystem masses listed below. These values are used directly for load derivation and local interface checks.

Subsystem	Mass (g)	Mass fraction (%)	axial load (N)	lateral load (N)
Prop	117549.40	34.50	9800	3460
Main Structure	112921.44	33.10	9420	3320
Avionics & Navigation	32959.42	9.70	2750	970
Instruments	2041.61	0.60	170	60
EPS	8184.62	2.40	680	240
Honeycomb	67000.00	19.60	5590	1970
Fasteners	5284.00	1.50	440	155
Total	341146.60	100	28450	10040

Table 2: Mass distribution

**The total in that table only represents the listed dry structural/subsystem mass of 341.15 kg*

The propulsion subsystem dominates the design-driving loads. With a mass of 117.5 kg, it alone produces approximately 9.8 kN axial and 3.46 kN lateral inertial load under the assumed launch case. This means the propulsion deck, tank mounts, and load path into the lower bus and interface must be treated as primary structure rather than secondary support hardware, as they carry a significant portion of the spacecraft inertial loading.

8 Structural Architecture and Intended Load Path

This section presents the detailed mechanical design of the spacecraft with a focus on the Structures and Mechanisms (SAM) subsystem. The design is described in terms of individual structural elements and subsystems, including their function, configuration, and performance. Emphasis is placed on how each component contributes to meeting the overall structural requirements, including load-bearing capability, subsystem integration, and compatibility with the launch vehicle.

The following assumptions were made during the development of the structural design:

- The spacecraft is subjected to 8.5 g axial and 3.0 g lateral loads during launch, based on Falcon 9 medium-class payload conditions.
- Structural behavior is assumed to be linear elastic for preliminary sizing.
- Load distribution across structural members is assumed to be uniform.
- Interfaces between subsystems are considered rigid in the preliminary stage.
- Thermal effects are considered qualitative but not fully modelled.
- Material properties are based on standard aerospace-grade values.

These assumptions enable initial sizing and validation of the structure, with further refinement planned through detailed simulation and testing.

9 Material Selection and Rationale

Material selection for the SAM subsystem is driven by the need to achieve high structural performance under launch loads while minimizing mass and ensuring manufacturability and environmental compatibility. The design adopts a hybrid material approach, using Ti-6Al-4V for primary load-bearing members and aluminium alloys with honeycomb cores for panels and secondary structures. This combination provides an optimal balance between strength, stiffness, weight, and cost, while maintaining compatibility with standard aerospace manufacturing practices.

Mixed-metal architecture is appropriate here because the highest loads are concentrated at interfaces, attachments, and fittings, whereas the larger-area panels and many beams benefit from aluminium's lower density and easier manufacturability.

Material	Typical use in this design	Density (kg/m ³)	Elastic modulus (GPa)	Yield strength (MPa)
Ti-6Al-4V	PAF attachment, Primary structure	4429	104.8	827
Al 7075-T6	Secondary structure	2810	72	505
Al 6061-T6	Secondary brackets, panel frames, SADA supports	2700	69	227
Al 5052(Face sheets)	Honeycomb panel face sheets	2680	70	90
HRH-10-3/16-5.4(Honeycomb core)	Sandwich panels core	72	0.015(effective)	2-3(core crush)

Table 3: Materials and their properties

Ti-6Al-4V is justified where local bearing, clamp, and interface stresses are severe, especially at the base interface and concentrated mounting lugs.

Aluminium alloys remain the most practical choice for large structural members and panels because they offer much lower mass penalty than titanium when volume rather than strength is governing, and they are easier to machine, inspect, and integrate with inserts and panel hardware.

9.1 Selection Criteria

Materials were selected based on the following key criteria:

- Strength-to-weight ratio: Ability to carry high loads with minimal mass
- Stiffness (E-modulus): To limit deflection and support pointing accuracy (SADA/HGA)
- Fatigue and durability: Resistance to cyclic loading during launch and operation
- Thermal stability: Minimal property degradation over -40°C to $+80^{\circ}\text{C}$
- Manufacturability: Compatibility with machining, bonding, and assembly processes
- Interface compatibility: Ability to join with dissimilar materials (bolted/bonded joints)
- Cost and availability: Use of aerospace-standard, readily available materials

9.1.1 Primary Structure – Ti-6Al-4V

Application: Main load-bearing frame (tubes, interface members, high-load brackets)

Key Properties (typical):

- **Elastic Modulus:** $1.048 \times 10^{11} \text{ N/m}^2$ ($\approx 104.8 \text{ GPa}$)
- **Poisson's Ratio:** 0.31
- **Yield Strength:** 827 MPa
- **Ultimate Tensile Strength:** 1050 MPa
- **Density:** 4429 kg/m³
- **Thermal Expansion Coefficient:** $9 \times 10^{-6} / \text{K}$

Rationale

The primary structure carries the majority of the spacecraft launch loads. Using the total wet mass of **1760.4 kg**, the Falcon 9 medium-class launch load case produces an axial compression load of:

$$F_{axial} = 1760.4 \times 8.5 \times 9.81$$
$$F_{axial} = 146,791 \text{ N} = 146.79 \text{ kN}$$

The lateral load is:

$$F_{lateral} = 1760.4 \times 3.0 \times 9.81$$
$$F_{lateral} = 51,809 \text{ N} = 51.81 \text{ kN}$$

Because the structure must safely carry approximately 146.79 kN axial load and 51.81 kN lateral load, Ti-6Al-4V was selected for the primary load-bearing members due to its high strength, stiffness, and fatigue resistance. Its yield strength of 827 MPa provides a large margin against yielding, while its elastic modulus of approximately 104.8 GPa improves stiffness and buckling resistance compared to aluminium alloys. This is especially important for the longerons and interface members, which must maintain structural stability under launch compression and lateral loading.

Design Implications

- a) Enables high load-carrying capability under the 146.79 kN axial launch load
- b) Improves buckling resistance of slender primary members
- c) Provides better fatigue performance under vibration and dynamic launch environments
- d) Supports reliable load transfer from decks and panels into the launch vehicle interface

Trade-offs

- i. Higher material and manufacturing cost than aluminium
- ii. More difficult machining compared to aluminium alloys
- iii. Requires careful interface design when joined to aluminium or CFRP components due to material compatibility and galvanic corrosion concerns.

9.1.2 Honeycomb Panel System – Aluminum Sandwich Panels

Application: External panels (side walls and top panel) providing structural enclosure and contributing to shear load distribution across the spacecraft.

Configuration

- i. **Face sheets:** Aluminium 5052, 0.8 mm thickness (each side)
- ii. **Core:** HRH-10-3/16-5.4 aluminium honeycomb, 19.8 mm thickness
- iii. **Total panel thickness:** 21.4 mm

Key Properties of Honeycomb Core **(Derived from Hexcel datasheets)*

- **Elastic Modulus:** 1.5×10^7 N/m²
- **Shear Modulus:** 6.0×10^6 N/m²
- **Density:** 72 kg/m³
- **Tensile Strength:** 2.0 MPa
- **Compressive Strength:** 3.0 MPa
- **Yield Strength:** 2.0 MPa
- **Thermal Expansion Coefficient:** 2×10^{-5} /K
- **Material Damping Ratio:** 0.03

Rationale

The honeycomb sandwich panels are designed to support the spacecraft under launch loads derived from the total wet mass of 1760.4 kg, corresponding to 146.79 kN axial and 51.81 kN lateral loads. While the primary Ti-6Al-4V structure carries the majority of axial loading, the honeycomb panels act as secondary structural elements, primarily resisting shear and out-of-plane bending.

The honeycomb core has a relatively low elastic modulus but provides high shear stiffness relative to its weight, which is critical for distributing lateral loads and stabilizing the structure. The 19.8 mm core thickness significantly increases the second moment of area, enabling the panel to behave as a stiff shear wall despite the low-density core.

Additionally, the material damping ratio (~0.03) improves vibration attenuation, which is beneficial for reducing dynamic response during launch.

Design Benefits

- Provides efficient shear load transfer under ~51.81 kN lateral loading
- Significantly increases panel bending stiffness with minimal mass
- Reduces overall spacecraft mass due to low core density (72 kg/m³)
- Improves vibration damping and dynamic stability
- Maintains structural integrity while supporting distributed subsystem loads

Trade-offs

- Low core strength ($\sim 2\text{--}3$ MPa) requires careful design of load introduction points
- Susceptible to core crushing and local failure near fasteners
- Requires potted inserts or reinforced interfaces for bolted connections
- More complex manufacturing due to bonding and curing processes

9.1.3 Secondary Structure – Aluminum Alloys (Al 7075-T6 / Al 6061-T4)

Application

- i. Propulsion deck and structural stiffeners
- ii. Mounting brackets and subsystem support
- iii. Interface plates and secondary load paths

Key Properties

- **Yield Strength:**
 - Al 7075-T6: ~ 505 MPa
 - Al 6061-T4: ~ 227 MPa
- **Elastic Modulus:** $\sim 69\text{--}72$ GPa
- **Density:** $\sim 2700\text{--}2810$ kg/m³
- **Good machinability and availability**

Rationale

Aluminium alloys are used for secondary structural components that support and distribute loads from subsystems into the primary structure. Under Falcon 9 medium-class launch conditions, the spacecraft experiences approximately 146.79 kN axial and 51.81 kN lateral loads, which are primarily carried by the Ti-6Al-4V primary structure.

Al 7075-T6 is used in higher-load secondary components such as the propulsion deck and structural beams due to its high strength, while Al 6061-T4 is used in lower-load brackets and support structures. Compared to titanium, aluminium alloys are easier to machine, more cost-effective, and sufficient for components that do not govern the primary load path.

Design Benefits

- a) Reduces cost and manufacturing complexity compared to titanium
- b) Provides sufficient strength for distributed subsystem loads
- c) Enables lightweight structural support for decks and mounting interfaces
- d) Good compatibility with aluminium honeycomb panel systems

Trade-offs

- i. Lower strength and stiffness compared to Ti-6Al-4V
- ii. Higher deflection under load due to lower modulus
- iii. Requires additional stiffening in some regions to maintain structural rigidity

9.1.4 Fasteners – A286 Stainless Steel

Application

- i. Structural joints
- ii. Subsystem mounting interfaces
- iii. Panel and deck attachments

Key Properties

- a) **High tensile strength:** ~900 MPa
- b) **Good temperature resistance** under launch and operational conditions
- c) **Corrosion resistant** with stable performance in aerospace environments

Rationale

A286 stainless steel is selected as the primary fastener material due to its high strength, reliability, and stability under thermal and mechanical loading. A286 fasteners ensure that loads are safely transmitted across connections without failure or loss of preload.

Additionally, A286 maintains mechanical properties over a wide temperature range, making it suitable for launch and space environments. Its compatibility with both Ti-6Al-4V and aluminium, allows for reliable multi-material integration.

Design Considerations

- i. Used for all critical structural joints and load-bearing connections
- ii. Enables reliable load transfer across interfaces under launch loading
- iii. Maintains preload under vibration and dynamic conditions
- iv. Compatible with both titanium and aluminium components, with appropriate corrosion mitigation

9.2 Material Compatibility and Interfaces

Since the design uses a combination of titanium and aluminium, material compatibility is an important consideration.

Key Considerations

- **Galvanic Corrosion Prevention**

Dissimilar material interfaces, particularly between titanium and aluminium, can lead to galvanic corrosion.

This is mitigated through the use of:

- i. Protective coatings (e.g., anodizing, conversion coatings)
- ii. Isolation washers or bushings
- iii. Sealants at contact interfaces

- **Thermal Expansion Mismatch**

Aluminium alloys ($\sim 2.36\text{--}2.4 \times 10^{-5}$ /K) expand more than Ti-6Al-4V ($\sim 9 \times 10^{-6}$ /K).

This mismatch is accounted for in the joint design by:

- i. Allowing for differential expansion in bolted interfaces
- ii. Avoiding rigid constraints that could induce thermal stresses
- iii. Using compliant interface features where required

- **Joint Design Strategy**

- i. Bolted joints (A286 fasteners) are used for primary interfaces to ensure reliable load transfer and maintain preload under vibration and dynamic loading
- ii. Welding between dissimilar materials is avoided due to incompatibility and residual stress concerns
- iii. Local reinforcement (inserts, backing plates) is used where loads are introduced into honeycomb panels

The combined material strategy ensures reliable structural performance, minimizes degradation due to environmental effects, and supports effective load transfer across all subsystem interfaces.

9.3 Environmental Considerations

The selected materials Ti-6Al-4V primary structure, Al 7075-T6 / Al 6061-T4 secondary structure, Al 5052 face sheets, HRH-10-3/16-5.4 honeycomb core, and A286 fasteners, are suitable for the expected space environment and launch conditions.

Key Considerations

- **Thermal Operating Range**

All materials are capable of operating within the required temperature range of -30°C to $+65^{\circ}\text{C}$ (SAM_07 requirement) while maintaining structural integrity. The selected alloys retain their mechanical properties within this range, ensuring reliable performance under launch and in-orbit conditions.

- **Vacuum Compatibility**

All selected materials are compatible with vacuum conditions, exhibiting low outgassing characteristics and stable performance in space environments.

- **Thermal Cycling and Structural Stability**

The materials maintain their mechanical properties under repeated thermal cycling, ensuring that the structure continues to safely carry launch loads of approximately 146.79 kN axial and 51.81 kN lateral without degradation.

Overall, the material selection ensures reliable performance under thermal, vacuum, and dynamic conditions, supporting both launch survivability and mission operations.

10 Spacecraft-Level Load Derivation

- **Total wet mass:** 1760.4 kg

- **Falcon 9 payload class:** Medium class, because it is between **1000–1800 kg**
- **Falcon 9 limit loads for medium class:**
 - Axial compression: **+8.5 g**
 - Axial tension: **−4.0 g**
 - Lateral: **±3.0 g**

The Falcon 9 User Guide lists these quasi-static load factors in Section 5.3.2 / Table 5-3, and notes that they are maximum flight limit loads, not qualification loads.

10.1 Load Calculations

10.1.1 Axial Compression Load

$$F_{axial} = m \cdot 8.5g$$

$$F_{axial} = 1760.4 \times 8.5 \times 9.81$$

$$F_{axial} = 146,791 \text{ N}$$

$$\boxed{F_{axial} = 146.79 \text{ kN}}$$

10.1.2 Axial Tension Load

$$F_{tension} = m \cdot 4.0g$$

$$F_{tension} = 1760.4 \times 4.0 \times 9.81$$

$$F_{tension} = 69,078 \text{ N}$$

$$\boxed{F_{tension} = 69.08 \text{ kN}}$$

10.1.3 Lateral Load

$$F_{lat} = m \cdot 3.0g$$

$$F_{lat} = 1760.4 \times 3.0 \times 9.81$$

$$F_{lat} = 51,809 \text{ N}$$

$$\boxed{F_{lat} = 51.81 \text{ kN}}$$

10.1.4 Design Loads with Safety Factors

Using:

- Yield FoS = **1.25**
- Ultimate FoS = **1.40**
- Yield Design Load

$$F_{yield} = 1.25 \times 146.79$$

$$\boxed{F_{yield} = 183.49 \text{ kN}}$$

- Ultimate Design Load

$$F_{ult} = 1.40 \times 146.79$$

$$\boxed{F_{ult} = 205.51 \text{ kN}}$$

10.2 Main Longerons Calculation

- i. Number of longerons: 8
- ii. Material: Al 7075-T6
- iii. Cross-section: 50 mm × 50 mm × 4 mm square tube
- iv. Length: 2050 mm
- v. Cross-Sectional Area

Outer dimension:

$$b = 50 \text{ mm}$$

Wall thickness:

$$t = 4 \text{ mm}$$

Inner dimension:

$$b_i = 50 - 2(4) = 42 \text{ mm}$$

$$A = b^2 - b_i^2$$

$$A = 50^2 - 42^2$$

$$A = 2500 - 1764$$

$$\boxed{A = 736 \text{ mm}^2}$$

10.2.1 Load Per Longerons

$$F_{member} = \frac{146.79}{8}$$

$$\boxed{F_{member} = 18.35 \text{ kN}}$$

At ultimate load:

$$F_{member,ult} = \frac{205.51}{8}$$

$$\boxed{F_{member,ult} = 25.69 \text{ kN}}$$

10.2.2 Direct Compressive Stress

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{18,349}{736 \times 10^{-6}}$$

$$\boxed{\sigma = 24.93 \text{ MPa}}$$

At ultimate design load:

$$\sigma_{ult} = \frac{25,689}{736 \times 10^{-6}}$$

$$\sigma_{ult} = 34.90 \text{ MPa}$$

For Al 7075-T6:

$$\sigma_y \approx 503 \text{ MPa}$$

$$FoS = \frac{503}{24.93}$$

$$FoS \approx 20.2$$

So, by direct compressive stress, the longerons are very safe.

10.2.3 Longerons Buckling Check

The second moment of area for the square hollow longeron is calculated as:

$$I = \frac{b^4 - b_i^4}{12}$$

$$I = \frac{50^4 - 42^4}{12}$$

$$I = 261,525 \text{ mm}^4 = 2.615 \times 10^{-7} \text{ m}^4$$

Euler Buckling Load

The critical buckling load is determined using Euler's formula:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Assuming:

- $E = 7.2 \times 10^{10} \text{ N/m}^2$ (Al 7075-T6)
- $L = 2.05 \text{ m}$
- $K = 1$ (pinned-pinned, conservative assumption)

$$P_{cr} = \frac{\pi^2 (7.2 \times 10^{10}) (2.615 \times 10^{-7})}{(2.05)^2}$$

$$P_{cr} \approx 44.2 \text{ kN}$$

Buckling Margin

At limit load per longeron:

$$MS = \frac{44.2}{18.35}$$

$$MS = 2.41$$

At ultimate load per longeron:

$$MS = \frac{44.2}{25.69}$$
$$\boxed{MS = 1.72}$$

The longerons are acceptable for preliminary sizing. While direct compressive stresses are well below the material yield strength, buckling is the governing failure mode for the slender members. The calculated margins indicate that the structure maintains adequate stability under both limit and ultimate loading conditions.

10.3 Propulsion Deck Calculation

1. Number of stiffener tubes: 4
2. Pattern: curved/cross-style layout around tank ring
3. Attachment: bolted
4. Load-carrying tubes to upper structure: 4
5. Tube size: 30 mm × 30 mm × 3 mm
6. Tube length: 1369.02 mm each
7. Material: Al 7075-T6

10.3.1 Load case used

Using fuel/Pressurant mass:

$$m = 1054.8 \text{ kg}$$
$$F_{axial} = 1054.8 \times 8.5 \times 9.81$$
$$\boxed{F_{axial} = 87.95 \text{ kN}}$$

For 4 load-carrying tubes:

$$F_{tube} = \frac{87.95}{4}$$
$$\boxed{F_{tube} = 21.99 \text{ kN}}$$

Direct compressive stress

Tube area:

$$A = 324 \text{ mm}^2 = 324 \times 10^{-6} \text{ m}^2$$
$$\sigma = \frac{21,989}{324 \times 10^{-6}}$$
$$\boxed{\sigma = 67.87 \text{ MPa}}$$

For Al 7075-T6:

$$\sigma_y = 505 \text{ MPa}$$

$$FoS_y = \frac{505}{67.87}$$

$$FoS_y = 7.44$$

So, by material yielding, the tube is safe.

10.3.2 Buckling check

For a $30 \times 30 \times 3$ mm hollow square tube:

$$I = \frac{30^4 - 24^4}{12}$$

$$I = 39,852 \text{ mm}^4$$

Using Euler buckling:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

With:

$$E = 72 \text{ GPa}$$

$$L = 1.369 \text{ m}$$

$$K = 1$$

$$P_{cr} = 15.11 \text{ kN}$$

Margin of Safety

$$MS = \frac{P_{cr}}{F_{tube}}$$

$$MS = \frac{15.11}{21.99}$$

$$MS = 0.69$$

Since:

$$MS < 1$$

The tube would buckle if treated as a fully unsupported pinned-pinned compression member.

10.4 Propulsion Tank / Ring Load Check

From mass table:

- PROP mature mass: 123.2 kg
- Fuel/Pressurant mass: 1054.8 kg
- Total wet spacecraft mass: 1760.4 kg

For the tank mount, the conservative case is to assume the tank ring supports the fuel/Pressurant mass.

- Fuel/Pressurant Axial Load

$$F = 1054.8 \times 8.5 \times 9.81$$

$$F = 87.92 \text{ kN}$$

- Lateral Load

$$F_{lat} = 1054.8 \times 3.0 \times 9.81$$

$$F_{lat} = 31.04 \text{ kN}$$

- Bolt Load

Consideration:

- Bolt count: 12
- Bolt size: M10
- Bolt circle diameter: 1071.6 mm
- Tank ring material: Ti-6Al-4V

Direct axial load per bolt:

$$F_{bolt} = \frac{87.92}{12}$$

$$F_{bolt} = 7.33 \text{ kN/bolt}$$

Direct lateral shear per bolt:

$$F_{shear} = \frac{31.04}{12}$$

$$F_{shear} = 2.59 \text{ kN/bolt}$$

This is acceptable as a preliminary load level for M10 bolts, but the final design still needs an overturning moment check once the tank CG height is known.

10.5 Instrument Deck Load Check

Deck Configuration:

- Deck type: CFRP honeycomb sandwich panel
- Deck size: 1500 mm × 1500 mm
- Face sheets: CFRP
- Face sheet thickness: 0.8 mm (each side)
- Core: HRH-10-3/16-5.4 aluminium honeycomb
- Core thickness: 19.8 mm

- Mounted instrument mass: 54.4 kg
- Attachment: Bolted to top structural assembly

Load Calculations

Axial Load (Launch Condition)

$$F = m \cdot n \cdot g = 54.4 \times 8.5 \times 9.81$$

$$F = 4.54 \text{ kN}$$

Lateral Load

$$F_{lat} = 54.4 \times 3.0 \times 9.81$$

$$F_{lat} = 1.60 \text{ kN}$$

Load Distribution at Mounting Points

If mounted with 4 attachment points:

$$F_{per\ point} = \frac{4.54}{4}$$

$$F_{per\ point} = 1.14 \text{ kN}$$

If mounted with 8 attachment points:

$$F_{per\ point} = \frac{4.54}{8}$$

$$F_{per\ point} = 0.57 \text{ kN}$$

Structural Considerations

Since the instrument deck is a CFRP honeycomb sandwich structure, the governing failure modes are not limited to global bending. The following local and panel-level failure mechanisms must be evaluated:

- Insert pull-out failure (critical for bolted joints)
- Face sheet bearing stress under fasteners
- Honeycomb core crushing under concentrated loads
- Local fasteners pull-through of CFRP face sheets
- Panel bending between supports due to distributed and point loads

10.6 Honeycomb Panel Mass Estimate

The total honeycomb panel surface area is:

$$A_{total} = 14.85 \text{ m}^2$$

Face sheet volume:

$$V_f = A_{total} \times t_f$$

$$V_f = 14.85 \times 0.0016 = 0.02376 \text{ m}^3$$

Using typical CFRP density:

$$\rho_{CFRP} \approx 1600 \text{ kg/m}^3$$

Face sheet mass:

$$m_f = \rho_{CFRP} \times V_f$$

$$m_f = 1600 \times 0.02376$$

$$\boxed{m_f = 38.02 \text{ kg}}$$

Honeycomb Core

For HRH-10-3/16-5.4:

$$\rho_{core} \approx 86.5 \text{ kg/m}^3$$

Core thickness:

$$t_c = 19.8 \text{ mm} = 0.0198 \text{ m}$$

Core volume:

$$V_c = A_{total} \times t_c = 14.85 \times 0.0198 = 0.29403 \text{ m}^3$$

Core mass:

$$m_c = \rho_{core} \times V_c = 86.5 \times 0.29403$$

$$\boxed{m_c = 25.43 \text{ kg}}$$

Total CFRP honeycomb panel mass:

$$m_{panel,total} = 38.02 + 25.43$$

$$\boxed{m_{panel,total} = 63.45 \text{ kg}}$$

With adhesive, edge closeouts, inserts, and fasteners:

$$\boxed{m_{panel,total,realistic} \approx 65 - 67 \text{ kg}}$$

10.7 PAF / Clampband Interface

From the Falcon 9 User Guide, SpaceX states that the standard mechanical interface is a 1575 mm diameter bolted interface at the forward end of the launch vehicle PAF.

The guide also states that SpaceX supports standard 937 mm, 1194 mm, and 1666 mm clampband interfaces, and that the specifications are provided in Appendix B.

For current design:

- PAF interface: 1575 mm
- Clamp band interface: 1194 mm
- Clamp band Circumference

$$C = \pi D$$

$$C = \pi \times 1.194$$

$$\boxed{C = 3.751 \text{ m}}$$

- Average Axial Line Load

$$q = \frac{F_{axial}}{C}$$

$$q = \frac{146,791}{3.751}$$

$$\boxed{q = 39.13 \text{ kN/m}}$$

This is the average axial load around the 1194 mm clampband. The actual local interface design will still be governed by:

- i. clampband groove shape
- ii. local flange stiffness
- iii. ring bending
- iv. separation system details
- v. bolt preload / clamp preload
- vi. local stress concentrations

11 Subsystem-Specific Structural Notes

The avionics subsystem has a mass of approximately 54.4 kg, resulting in an inertial load of approximately 4.54 kN under 8.5 g axial launch loading. While avionics enclosures are typically manufactured from aluminium alloys, their mounting interfaces including rails, brackets, and panel inserts must be carefully designed to ensure sufficient stiffness and to prevent fastener pull-through or local panel failure, especially when mounted on CFRP or honeycomb panels.

The instruments and payload components have relatively low mass but are highly sensitive to deflection and misalignment. As a result, their support structure should be connected through short, stiff load paths directly

into the primary structure, rather than relying on thin panel sections, to maintain alignment and functional performance during launch and operation.

The Electrical Power System (EPS), with a mass of approximately 56.7 kg, generates an axial load of approximately 4.72 kN under launch conditions. Critical interfaces such as the Solar Array Drive Assembly (SADA), support brackets, and hinge interfaces must be designed not only for launch loads but also for deployment loads, stop impacts, and torsional stiffness, which can govern local structural design.

Thermal subsystem components, with a mass of approximately 27.3 kg, produce an axial load of approximately 2.28 kN during launch. Although these components are relatively lightweight, elements such as radiators and panels often have large unsupported surface areas, requiring verification of edge support conditions, panel stiffness, and out-of-plane bending behaviour.

The SAM subsystem includes the primary structure, decks, and structural interfaces, which collectively carry a significant portion of the total spacecraft load derived from the 1760.4 kg wet mass. The spacecraft interfaces with the launch vehicle through a 1575 mm PAF and 1194 mm clampband system, which defines the primary load transfer path. The base interface structure must therefore be designed to efficiently transfer the global launch loads of approximately 146.79 kN axial into the primary structure while minimizing stress concentrations.

12 Future Work

The current design represents a preliminary structural configuration that satisfies initial sizing and load requirements. However, further refinement, validation, and integration are required to progress toward a fully flight-ready design. The following activities outline the key areas of future work for the Structures and Mechanisms (SAM) subsystem.

12.1 Refinement of Structural Design and Analysis

The structural design will be further refined through detailed analysis and validation. While preliminary calculations have been performed using conservative assumptions, higher-fidelity methods are required to capture real structural behaviour.

Future work includes:

- Development of detailed Finite Element Analysis (FEA) models of the full spacecraft structure
- Evaluation of static loads, modal characteristics, and vibration response
- Verification of buckling behaviour for longerons and slender members
- Analysis of local stresses at critical interfaces (deck supports, tank mounts, fasteners)
- Correlation of analytical results with simplified hand calculations

This step will ensure that the structure meets all launch vehicle requirements and that no critical failure modes are overlooked.

12.2 Mass Refinement and Margin Tracking

The current mass estimates are based on preliminary CAD models and assumptions. As the design evolves, mass properties will be continuously updated and tracked.

Future work includes:

- Integration of updated mass properties from all subsystems
- Tracking of mass growth and system-level margins
- Refinement of component-level mass using detailed CAD and material definitions
- Identification of mass optimization opportunities, particularly in over-designed structural members

Maintaining mass margins is critical to ensuring compatibility with launch vehicle constraints.

12.3 Finalization of Materials and Manufacturing Approach

Material selection has been defined at a preliminary level; however, final decisions must consider manufacturability, cost, and performance.

Future work includes:

- Down-selection and validation of final material specifications
- Development of detailed CAD models with manufacturing-level features
- Creation of engineering drawings with GD&T specifications
- Evaluation of manufacturing processes (machining, bonding, assembly methods)
- Design of interfaces between dissimilar materials (Ti, Al, CFRP)

This stage ensures that the design can be reliably manufactured and assembled.

12.4 Mechanisms and Interface Development

Future work includes:

- Detailed design and integration of the Solar Array Drive Assembly (SADA)
- Development of hinges, deployment mechanisms, and hold-down release mechanisms (HDRMs)
- Finalization of subsystem mounting interfaces (avionics, propulsion, payload)
- Verification of load transfer across interfaces
- Integration of mechanical systems within the overall spacecraft architecture

This step ensures that all moving and interface components function reliably under both launch and operational conditions.

12.5 Testing Plan and CDR Readiness

A comprehensive testing and validation strategy will be developed to support the transition to Critical Design Review (CDR).

Future work includes:

- Definition of structural validation tests (static load testing, vibration testing)
- Development of test plans and acceptance criteria
- Identification of critical components requiring qualification testing
- Preparation of verification matrices linking requirements to tests and analysis
- Documentation and review preparation for CDR readiness

This phase ensures that the design is fully validated and meets all system requirements before final approval.

12.6 Risk Reduction and Design Optimization

In addition to validation, further work will focus on reducing design risks and improving overall system efficiency.

Future work includes:

- Identification and mitigation of structural risks (buckling, local failure modes, interface loads)
- Optimization of structural mass while maintaining safety margins
- Refinement of load paths to eliminate unnecessary stress concentrations

13 Lessons Learned

The development of the Structures and Mechanisms (SAM) subsystem provided several key insights into spacecraft structural design, subsystem integration, and design process management. These lessons are critical for improving both the current design and future projects.

13.1 SAM as an Iterative Subsystem

The structural subsystem is inherently iterative due to its strong coupling with other subsystems, particularly propulsion and system-level mission requirements. Changes in mission parameters, such as ΔV requirements, directly impact propellant mass, which in turn affects the total spacecraft mass and structural loading

13.2 Role of Heritage in Design Efficiency

The use of heritage components plays a significant role in reducing design complexity and development time. While heritage is often associated with reliability, its primary advantage lies in minimizing the need for requalification and redesign.

13.3 Designing for Adaptability

Spacecraft design is dynamic and continues to evolve until final system integration. Throughout the design process, changes in mass, volume, and subsystem requirements are expected.

13.4 Importance of Maintaining Design Margins

Maintaining adequate design margins from the early stages is essential for managing uncertainty and accommodating future changes. Margins in mass, volume, and structural capacity provide necessary flexibility as the design matures.

14 SAM Subsystem Risks

The following key risks have been identified for the Structures and Mechanisms (SAM) subsystem based on the current preliminary design:

- **Buckling of Slender Members:**
Long structural members (longerons and support tubes) are susceptible to buckling under axial loads, which governs the design rather than material strength.
- **Propulsion Deck Load Concentration:**
High loads from the propellant tank are concentrated at the deck interface, creating risk of local deformation, fastener failure, or panel overstress.
- **Interface Load Transfer Uncertainty:**
Incomplete definition of load paths between subsystems (propulsion, avionics, payload) may lead to inaccurate stress distribution and potential structural issues.
- **Mass Growth Sensitivity:**
Increases in subsystem mass, particularly propulsion, directly increase structural loads and may require redesign.
- **Material Compatibility Risks (Ti–Al–CFRP):**
Mixed material interfaces introduce risks of galvanic corrosion, thermal expansion mismatch, and joint degradation.
- **Manufacturing and Assembly Complexity:**
Complex geometry and multi-material interfaces increase the risk of tolerance issues, misalignment, and assembly challenges.

15 Conclusion

This report presented the preliminary design and structural evaluation of the spacecraft Structures and Mechanisms (SAM) subsystem, developed to meet launch vehicle constraints, subsystem integration requirements, and mission objectives. The design establishes a robust structural architecture capable of supporting all onboard systems while maintaining efficiency in mass, stiffness, and manufacturability.

The spacecraft structure adopts a hybrid load-bearing approach, combining high-strength primary members with lightweight sandwich panel systems. The primary frame, decks, and interface structures collectively define a clear and efficient load path from subsystem components to the launch vehicle interface. The propulsion subsystem, identified as the dominant load driver, has been integrated through a reinforced deck and support structure to ensure reliable load transfer under launch conditions.

Using a total wet mass of 1760.4 kg, the structure was sized for Falcon 9 medium-class quasi-static loads, resulting in 146.79 kN axial and 51.81 kN lateral design loads. Analytical results indicate that the primary structural members operate well below material yield limits, with factors of safety exceeding acceptable thresholds. Buckling analysis identified stability as the governing design criterion, and preliminary margins demonstrate adequate structural integrity under both limit and ultimate loading conditions. Local interface checks, including propulsion tank mounting and deck support structures, also show acceptable performance at the preliminary design level.

The design further satisfies key system requirements, including launch vehicle interface compatibility, fairing envelope constraints, subsystem mounting provisions, and structural support for mechanisms such as the Solar Array Drive Assembly and High Gain Antenna. Where applicable, requirements have been met at a preliminary level, with identified areas requiring further validation through detailed analysis and testing.

Material selection has been carefully considered to balance structural performance and manufacturability. The use of titanium for high-load interfaces, aluminium alloys for secondary structures, and honeycomb sandwich panels for enclosure and shear load distribution provides an efficient and practical solution aligned with aerospace standards. Additionally, Design-for-X considerations have been incorporated to support manufacturability, assembly, and system reliability.

Overall, the SAM subsystem design demonstrates a structurally sound and feasible configuration that meets performance requirements with appropriate safety margins. The analysis confirms that the spacecraft structure can withstand expected launch loads and supporting mission operations. However, as this is a preliminary design, further refinement through high-fidelity analysis, detailed interface design, and structural testing is required to validate assumptions and progress toward Critical Design Review (CDR).

16 Appendices A: Subsystem Load Calculations

16.1 A.1 Methodology

The axial and lateral inertial loads for each subsystem are calculated using:

$$F = m \cdot n \cdot g$$

Where:

- F = Load (N)
- m = Mass (kg)
- n = Load factor (g-level)
- $g = 9.81 \text{ m/s}^2$

For Falcon 9 (medium-class payload):

- Axial load factor: **8.5 g**
- Lateral load factor: **3.0 g**

Since subsystem masses are given in grams:

$$m_{kg} = \frac{m_g}{1000}$$

Thus:

$$F_{axial} = \frac{m_g}{1000} \times 8.5 \times 9.81$$

$$F_{lateral} = \frac{m_g}{1000} \times 3.0 \times 9.81$$

16.2 A.2 Subsystem Calculations

16.2.1 Propulsion Subsystem

$$m = \frac{117549.4}{1000} = 117.55 \text{ kg}$$

$$F_{axial} = 117.55 \times 8.5 \times 9.81 = 9801 \text{ N}$$

$$F_{lateral} = 117.55 \times 3.0 \times 9.81 = 3459 \text{ N}$$

16.2.2 Main Structure

$$m = \frac{112921.44}{1000} = 112.92 \text{ kg}$$

$$F_{axial} = 112.92 \times 8.5 \times 9.81 = 9413 \text{ N}$$

$$F_{lateral} = 112.92 \times 3.0 \times 9.81 = 3322 \text{ N}$$

16.2.3 Avionics & Navigation

$$m = \frac{32959.42}{1000} = 32.96 \text{ kg}$$

$$F_{axial} = 32.96 \times 8.5 \times 9.81 = 2748 \text{ N}$$

$$F_{lateral} = 32.96 \times 3.0 \times 9.81 = 970 \text{ N}$$

16.2.4 Instruments

$$m = \frac{2041.61}{1000} = 2.04 \text{ kg}$$

$$F_{axial} = 2.04 \times 8.5 \times 9.81 = 170 \text{ N}$$

$$F_{lateral} = 2.04 \times 3.0 \times 9.81 = 60 \text{ N}$$

16.2.5 EPS (Electrical Power System)

$$m = \frac{8184.62}{1000} = 8.18 \text{ kg}$$

$$F_{axial} = 8.18 \times 8.5 \times 9.81 = 682 \text{ N}$$

$$F_{lateral} = 8.18 \times 3.0 \times 9.81 = 241 \text{ N}$$

16.2.6 Honeycomb Panels

$$m = \frac{67000}{1000} = 67.00 \text{ kg}$$

$$F_{axial} = 67.00 \times 8.5 \times 9.81 = 5590 \text{ N}$$

$$F_{lateral} = 67.00 \times 3.0 \times 9.81 = 1971 \text{ N}$$

16.2.7 Fasteners

$$m = \frac{5284}{1000} = 5.28 \text{ kg}$$

$$F_{axial} = 5.28 \times 8.5 \times 9.81 = 440 \text{ N}$$

$$F_{lateral} = 5.28 \times 3.0 \times 9.81 = 155 \text{ N}$$

16.3 A.3 Total Load Check (Listed Subsystems Only)

Total mass:

$$m = \frac{341146.6}{1000} = 341.15 \text{ kg}$$

$$F_{axial} = 341.15 \times 8.5 \times 9.81 = 28,450 \text{ N}$$

$$F_{lateral} = 341.15 \times 3.0 \times 9.81 = 10,040 \text{ N}$$

16.4 Important Note

The above calculations are based on the listed subsystem masses (**341.15 kg**), which represent a subset of the spacecraft. The full spacecraft wet mass is:

$$m_{total} = 1760.4 \text{ kg}$$

Corresponding total launch loads:

$$F_{axial,total} = 1760.4 \times 8.5 \times 9.81 = 146.79 \text{ kN}$$

$$F_{lateral,total} = 1760.4 \times 3.0 \times 9.81 = 51.81 \text{ kN}$$

These total loads are used for global structural sizing, while subsystem loads are used for local interface design.